MAHATMA EDUCATION SOCIETY'S

PILLAI COLLEGE OF ARTS, COMMERCE & SCIENCE
(Empowered Autonomous)

NEW PANVEL

PROJECT REPORT ON

Talent AI: Smart Resume screening and ranking system"

IN PARTIAL FULFILLMENT OF

M.Sc. DATA ANALYTICS

SEMESTER I– 2026-27

PROJECT GUIDE

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Talent AI: Smart Resume Screening and Ranking System

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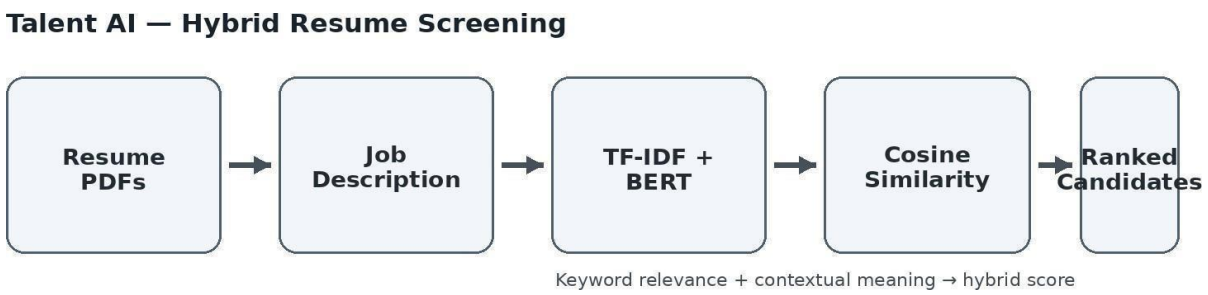
1. Abstract

The process of recruitment is a highly vital process for any organization, and the selection of the right person is directly proportional to the success of the company. But in the modern digital world, there are thousands of resumes sent for one particular job position. Not only is it a time-consuming process, but the process is inefficient. The traditional process of resume screening involves the process of matching keywords. But the process of matching keywords does not allow the process to understand the meaning of the skills possessed by the candidate or the meaning of the skills that need to be possessed for the job. In order to resolve the aforementioned issues, the present project is based on the development of "Talent AI," an intelligent resume screening and ranking system using AI technology. The proposed system is based on the use of the BERT model, along with TF-IDF. The proposed system converts the resume and the job position into numerical data points and compares them with each other to determine the similarity level.

In addition, a disk caching method is applied to store the computed data, which facilitates the enhancement of the system's speed and minimizes processing time. Resume processing, similarity, ranking, and performance evaluation are included in the system. The results indicate the efficiency of the Talent AI in resume screening, reducing the need for human involvement. This project shows the effectiveness of AI in modern recruitment systems.

2.Keywords

Talent AI, Resume Screening, Candidate Ranking, Semantic Matching, BERT, TF-IDF, AI, NLP, Recruitment Automation



2. Introduction

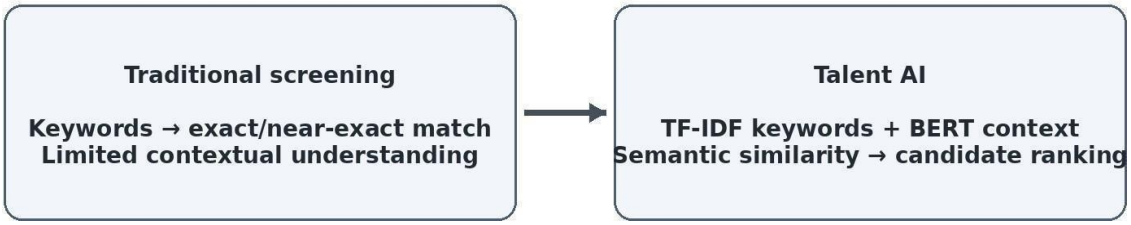
The recruitment process is regarded as one of the most important functions for any organization, as it affects the efficiency and productivity of the particular organization. In the recent past, because of the emergence of a number of online job sites, the number of resumes received by an organization has increased manifold. An organization might receive hundreds and thousands of resumes for a single vacancy announcement. It is a challenging task to evaluate such a huge number of resumes.

To avoid such challenges, it is better to use automated resume screening techniques. The traditional method of resume screening is based on the use of keyword matching techniques. The automated screening method is based on the comparison of the keywords available with the candidate's resume with the keywords available with the job description. This method is useful for screening a huge number of resumes. But the automated screening method based on the use of keyword matching techniques has a major drawback. The automated screening method based on the use of keyword matching techniques is not capable of understanding the actual meaning of the data available with the resume. For example, the resume of the candidate may contain the keywords "developed machine learning models," but the job description may contain the keywords "AI development skills." Both the keywords are similar, but the automated screening method is not capable of understanding the similarity between the keywords.

With the recent developments in the field of AI, the machines are now capable of understanding the human language. With the help of deep learning techniques like BERT, the machines are now capable of understanding the context of the text.The current study aims to develop an intelligent system named Talent AI, which uses semantic analysis with the help of BERT and keyword analysis with the help of TF-IDF. The proposed system will represent the resume and the job description as a numerical value, thereby calculating the similarity score to rank the candidates.

The main aim of the current study is to develop an intelligent system using AI techniques to screen the resumes of the candidates. The system should be capable of ranking the candidates based on the relevance of the resume to the job description. The system should also be capable of evaluating the performance of the model. The current study aims to show the importance of AI in the current recruitment process.

From Keyword Matching to Semantic Matching



4.Literature review

Several studies have been conducted to explore different ways of automatically screening resumes by employing machine learning and natural language processing techniques. Traditional keyword matching algorithms and cosine similarity algorithms have been used for resume screening. However, these methods have limitations in understanding the contextual meaning of skills and qualifications. Deep learning algorithms such as LSTM, CNN-GRU, and neural networks have also been explored for resume analysis and skill identification, but robust candidate ranking and contextual understanding remain important challenges.

Recent research has increasingly focused on semantic job-resume matching. A 2026 study on AI-driven semantic similarity-based job matching proposed a recruitment framework using TF-IDF vectorization, cosine similarity, and domain-specific keyword weighting. The study reported that semantic matching can improve job-candidate alignment beyond exact keyword matching and can capture related skills and contextual relevance.

Another 2026 study developed a CV screening system using BERT, Gemini, and TF-IDF. The research compared these approaches for candidate evaluation and found that transformer and large-language-model approaches can provide stronger semantic understanding, while TF-IDF remains useful for identifying important lexical terms. This supports the use of hybrid approaches that combine keyword-level information with contextual language representations.

A 2026 systematic review of artificial intelligence in recruitment and personnel selection reported that AI and NLP are increasingly used for candidate pre-selection, resume classification, job suitability prediction, and personalized candidate recommendations. The review also identified challenges such as non-standardized resume formats, linguistic variability, limited data, interpretability, and the possibility of excluding atypical candidate profiles.

Recent work has also emphasized explainable candidate ranking. Research on explainable resume screening has explored semantic embeddings, Sentence-BERT, attention-based ranking, efficient similarity search, and explanation methods such as SHAP and LIME. These approaches aim to make automated screening more transparent by showing the factors that influence candidate recommendations and rankings.



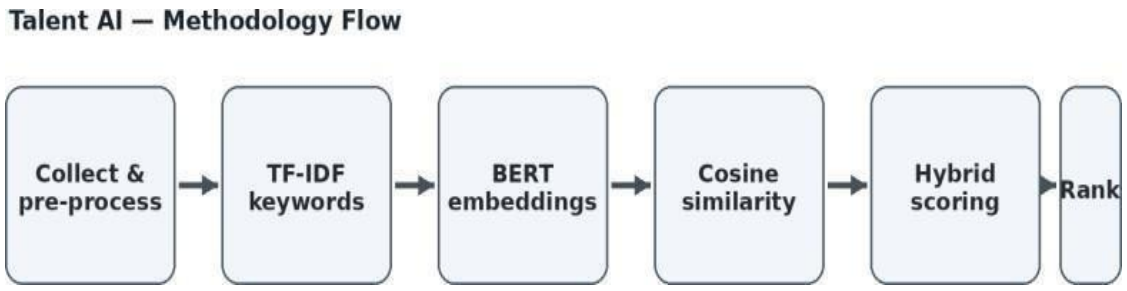
Fairness and algorithmic bias have become important concerns in AI-based recruitment. A 2026 study on adversarial contrastive tuning for LLM-based resume screening investigated methods for reducing bias while maintaining screening performance. Other 2026 research has also examined bias in AI hiring systems and emphasized the importance of fairness evaluation, human review, and responsible deployment. Therefore, modern resume screening systems should focus not only on accuracy and efficiency but also on transparency and fair treatment of candidates.

Recent literature on AI-based job matching also indicates that data-driven systems are scalable but may suffer from dataset bias and low interpretability. Hybrid approaches that combine different representations of jobs, skills, and candidates are therefore becoming an important research direction. This supports the development of a system that combines lexical matching with semantic matching rather than depending on only one technique.

Based on the reviewed literature, there is scope for an integrated resume screening system that combines keyword relevance, semantic similarity, candidate ranking, and explainable results. The proposed Talent AI system addresses this direction by combining BERT and TF-IDF. TF-IDF is used to identify important keywords, while BERT is used to capture contextual meaning. Cosine similarity is then used to compare resumes with job descriptions, and a hybrid scoring mechanism combines the two results. This approach is consistent with the recent research direction toward semantic, efficient, explainable, and responsible AI-assisted recruitment systems.

5.Methodology

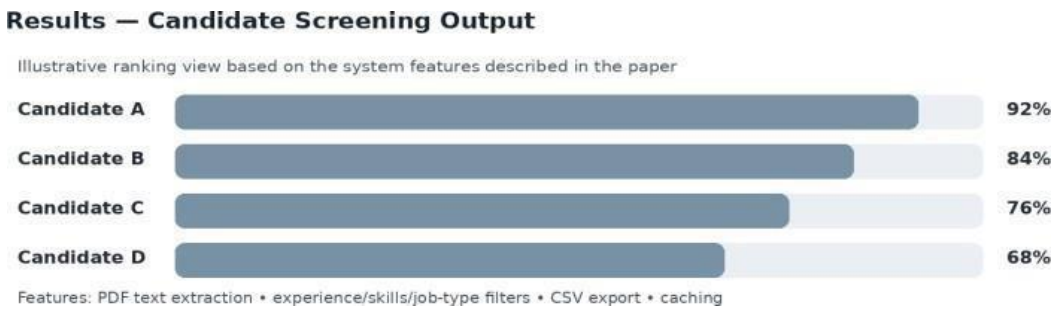
The proposed system for Talent AI is intended to incorporate a systematic approach in order to automate the resume screening and ranking of candidates. In order to achieve this, the resume data is first collected and pre-processed to clean the text and make it lowercase, and the skills and resume summaries are merged accordingly. Once the resume data is pre-processed, two different methods are employed to perform similarity analysis on the resumes and job descriptions. The TF-IDF model is employed to identify the key words in the resume as well as the job description, and the BERT model is employed to produce semantic embeddings to interpret the context of the data. The cosine similarity measure is employed to determine the similarity between the resumes and job descriptions. In order to achieve a high level of accuracy in the outcome, a hybrid scoring system is proposed to merge the outcome of the BERT model's score and the TF-IDF model's score. The proposed system architecture incorporates the use of a frontend interface, a backend, and an AI module to execute the similarity analysis on the resumes and job descriptions.



6. Results

The Talent AI system analyzed the job descriptions using a combined approach. This approach used both BERT and TF-IDF methods to evaluate and rank candidates based on how well they matched the job requirements. The hybrid model showed the best results for accurately ranking the candidates. The application demonstrated efficient data processing capabilities, leveraging a disk caching system for BERT embeddings; consequently, the system's initialization time following the initial execution was roughly 10 seconds. Furthermore, the system effectively handled resume screening, text extraction from PDF documents, and candidate ranking predicated on their alignment with the specified job description. The system now includes advanced features, such as filters for experience, skills, and job type. The new system boasts sophisticated features, including filters that allow users

to sort candidates by experience, skills, and the specific type of job they're applying for. Exporting the ranked results as CSV files made it easy to share and analyze the data later. The automated resume screening and candidate ranking capabilities proved their worth, leading to a marked increase in the recruitment process's overall efficiency.



7.Conclusion

Talent AI presents a smart resume screening system through a hybrid semantic matching approach. The recruitment process is made more accurate and efficient. By using BERT and TF-IDF in the resume screening system, we can achieve a better understanding of the text and a more precise ranking of candidates.

8.Acknowledgement

I am very grateful to my project guide and the faculty at Pillai College of Arts, Commerce and Science. Their support was essential to the success of this project, which is called 'Talent AI: Smart Resume Screening and Ranking System. I am grateful to them for their time and energy in mentoring and guiding me.

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